Modeling Monopoly as a Markov Chain: Long-Run Landing Probabilities Under Official Rules

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Abstract

Although Monopoly appears to be governed primarily by chance, its rules introduce structural constraints that bias player movement across the board. In this project, I model Monopoly as a Markov chain and compute the stationary distribution of landing probabilities under official gameplay rules. By expanding the state space to incorporate doubles streaks and jail states, the model preserves the Markov property and yields an exact theoretical distribution rather than a simulation-based approximation. The results reveal strong positional asymmetries, including a disproportionately high probability of occupying Jail and elevated traffic on squares following Jail. These findings demonstrate how deterministic rules can induce predictable long-run behavior.

1. Introduction

Monopoly is widely perceived as a game driven purely by chance. However, players often develop strong intuitions about which properties are visited more frequently or which areas of the board feel dangerous. Such intuitions raise a natural question. Are these patterns real, or are they simply cognitive biases?

This project began from a personal observation: when playing Monopoly, I appeared to spend a disproportionate amount of time in jail. Rather than attributing this to bad luck, I investigated whether the rules of the game mathematically favor certain outcomes. To answer this question, I constructed a probabilistic model of Monopoly using Markov chains and computed the exact long-run probability of landing on each square.

2. Modeling Framework

2.1. Limitations of a Naïve Model

A simple model might treat each of the 40 board squares as a state. However, such a representation is not truly Markovian. In Monopoly, the probability of a player's next move depends not only on their current position but also on hidden variables such as the number of consecutive doubles rolled and whether the player is currently in jail. Because transition probabilities depend on past events, the process is not memoryless when defined solely in terms of board position.

2.2. State Space Design

To preserve the Markov property, the state must encode all information relevant to future transitions. States were therefore defined as

$$(\text{position}, \text{doubles streak})$$

along with additional jail states representing successive attempts to leave jail. This expanded state space ensures that

$$P(X_{t+1} \mid X_t, X_{t-1}, \ldots) = P(X_{t+1} \mid X_t),$$

making the process a valid Markov chain.

State Indexing Convention For clarity, we specify the board indexing convention used throughout this paper. Board positions are represented numerically from 0 to 39, with 0 denoting GO and indices increasing clockwise until 39, which corresponds to Boardwalk.

3. Transition Structure

The transition matrix incorporates all official movement rules, including dice probabilities, rolling doubles, three consecutive doubles sending a player to jail, Chance card instructions, Community Chest card instructions, the "Go to Jail" square, and rules governing release from jail. Each dice outcome contributes probabilistically weighted transitions, while card instructions introduce deterministic jumps to other states.

4. Stationary Distribution

The long-run behavior of a Markov chain is characterized by its stationary distribution π, which satisfies

$$\pi P = \pi.$$

Rather than solving this equation analytically, the stationary distribution was computed numerically through iterative multiplication,

$$\pi_{t+1} = \pi_t P,$$

which converges to a unique solution because the chain is irreducible and aperiodic under standard Monopoly rules. The resulting vector π represents the fraction of time a player spends on each square over infinitely many turns.

5. Results

The stationary distribution reveals several structural properties of the game. Jail is the most frequently occupied state by a large margin because multiple rules direct players there and players remain there for several turns. Squares immediately following Jail receive elevated traffic because many turns begin from that location. Illinois Avenue is among the most frequently landed-on

properties, benefiting both from its position and from card instructions directing players there. In contrast, Chance squares have low stationary probability because players rarely remain on them after landing. The "Go to Jail" square likewise exhibits very low probability because players are immediately moved elsewhere upon arrival.

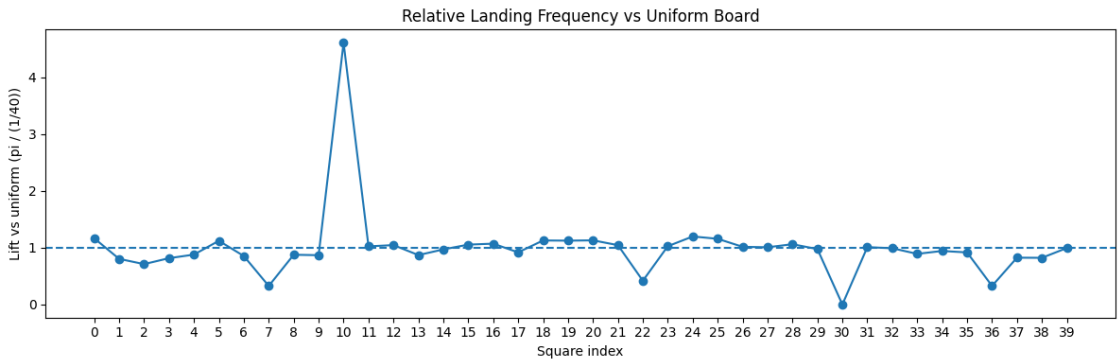

Figure 1: Relative landing frequency compared to a uniform distribution (1/40 baseline). Jail (10) exhibits a disproportionately high stationary probability.

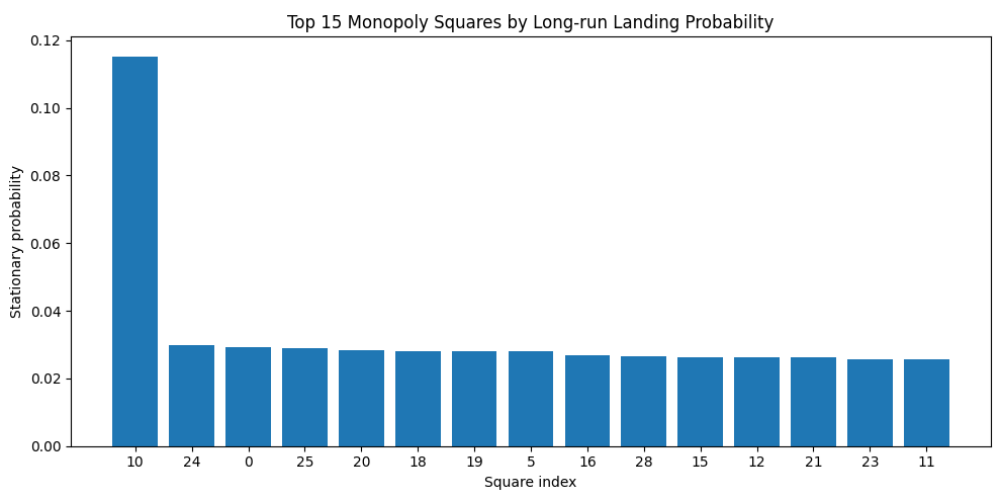

Figure 2: Top 15 squares ranked by stationary landing probability. Jail dominates, masking the variation between other spaces.

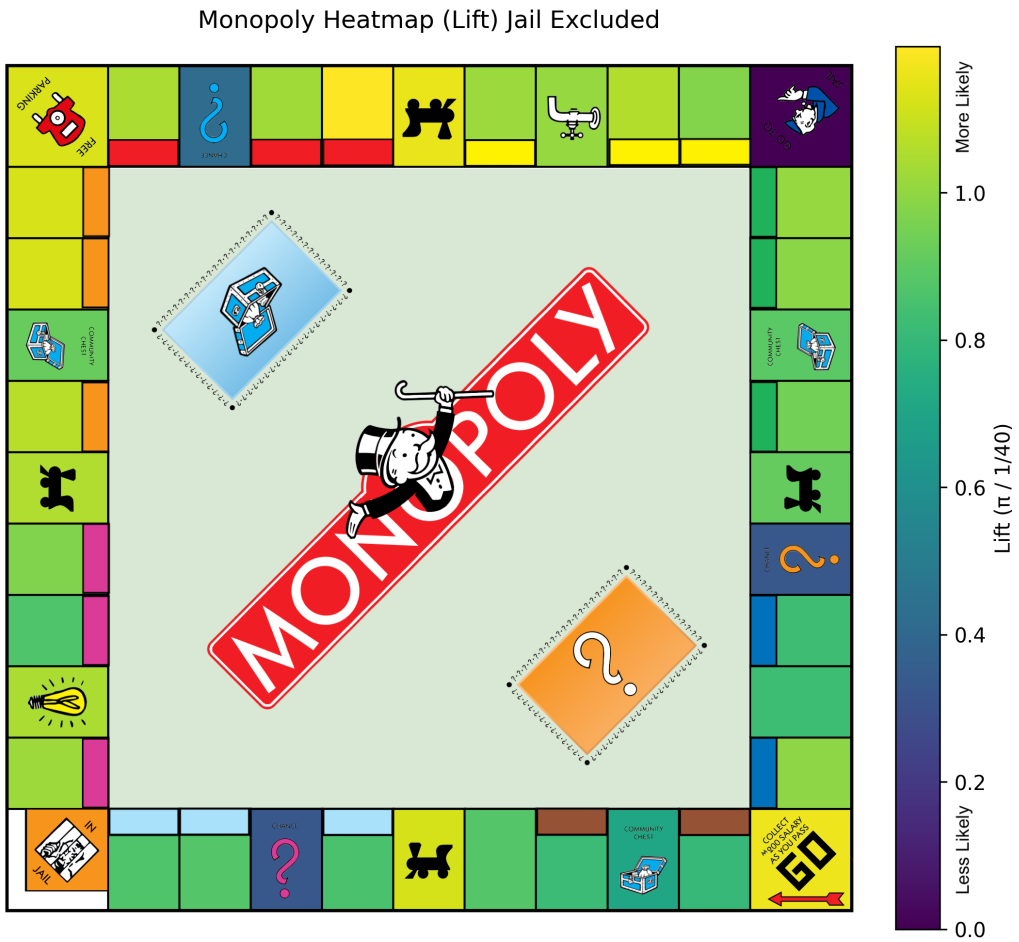

Figure 3: Heatmap with Jail excluded to reveal finer variation among remaining squares.

6. Interpretation

These results demonstrate that Monopoly's board is not probabilistically symmetric, even though it appears geometrically symmetric. The rules of the game introduce directional biases that systematically shape player movement. In particular, Jail acts as a probability sink, while squares immediately following it function as probability accumulation zones. More broadly, this illustrates a general principle in stochastic systems: long-run probability depends on residence time, not merely visit frequency.

7. Conclusion

Although Monopoly is commonly viewed as a game of chance, its rules impose a statistical structure that strongly influences long-run outcomes. By modeling the game as a Markov chain and computing its stationary distribution, this project reveals the underlying probabilistic dynamics governing player movement. What initially appeared to be anecdotal experience turns out to be supported by mathematical analysis. The game is not purely random. It is a stochastic system

shaped by deterministic rules.